



GS FOUNDATION 2024-25

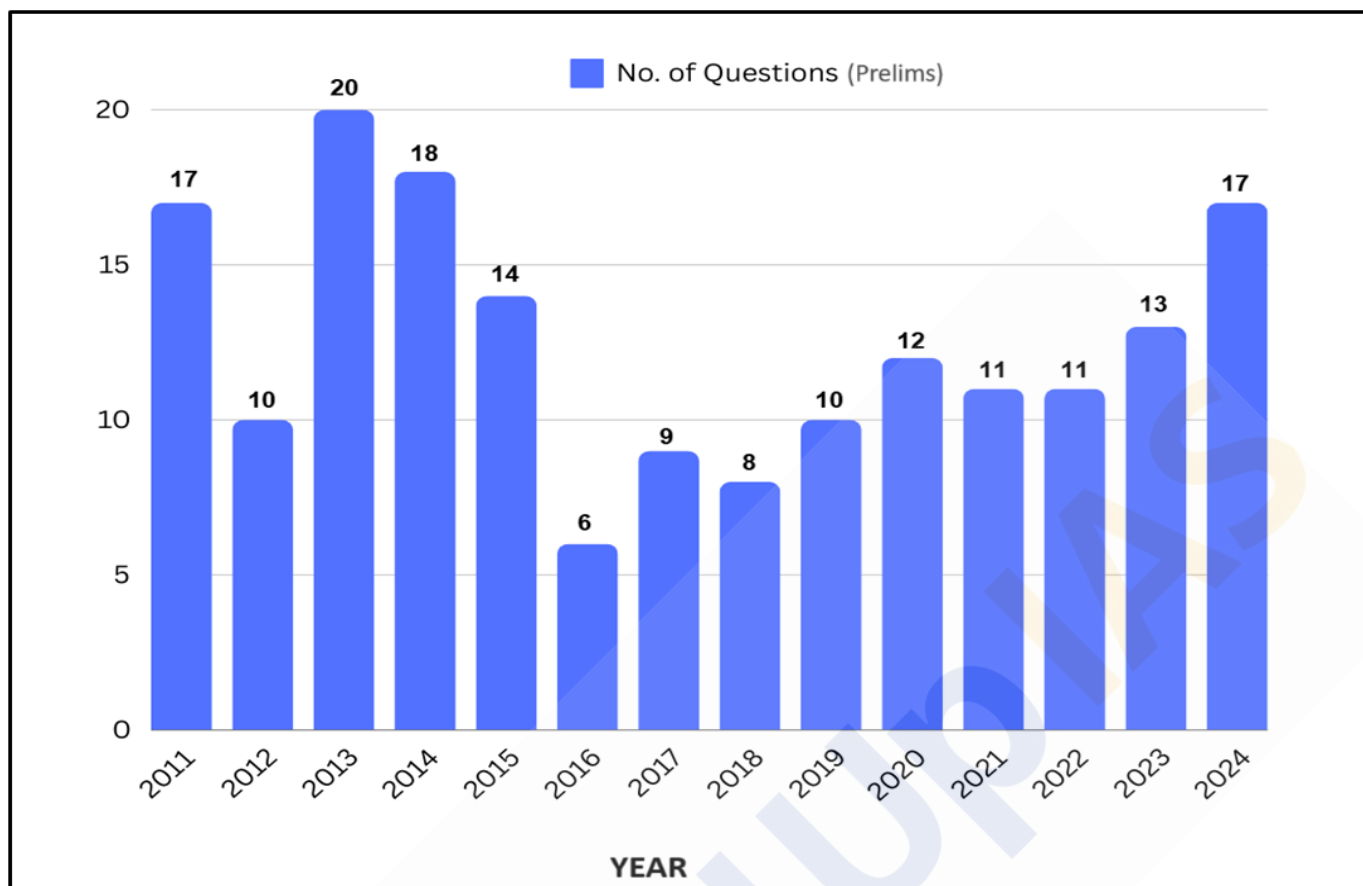
GEOGRAPHY - 1

ORIENTATION & BASICS OF GEOGRAPHY

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Basics of Geography

Introduction:

The word 'Geography' has been derived from the Greek words 'Geo' meaning the **earth** and 'Graphos' meaning to **write or describe**. In very simple words, it can be said that geography is the **description of the earth**. The term geography was first coined by **Eratosthenes**, a Greek scholar (276-194 BC.). He is also known as the "**Father of Geography**".



What does Geography study?

- Geography is the study of **Earth's landscapes, peoples, places and environments**. It is, quite simply, about the world in which we live.
- Geography is unique in bridging the **social sciences (human geography)** with the **natural sciences (physical geography)**.
- It also studies spatial organization and spatial integration.

Geography as a discipline is concerned with 3 sets of questions:

1. **Identification:** Some questions are related to the **identification of the patterns of natural and cultural features** as found over the surface of the earth. These are the questions about **What?**
2. **Distribution:** Some questions are related to the **distribution of the natural and human/cultural features** over the surface of the earth. These are the questions about **Where?**

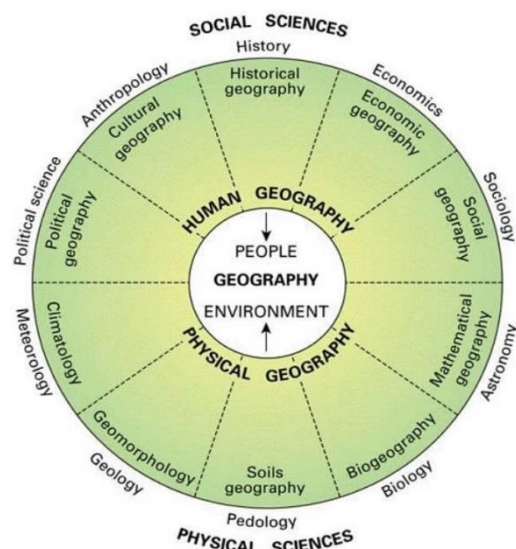
Taken together, both these questions take care of distributional and locational aspects of the natural and cultural features. But, these two questions did not make geography a scientific discipline until the third question was added.

3. **Explanation:** The third question is related to the explanation or the **causal relationships between features, processes, and phenomena**. This aspect of geography is related to the question, **Why?**

Geography as an integrating discipline:

Geography is a **discipline of synthesis** and attempts **spatial synthesis**. Its approach is holistic in nature and recognises the fact that the world as a **system of interdependencies, integrating natural and social sciences** to understand reality in its spatial perspective.

Geography influences historical events through spatial factors, such as in India, where the Himalayas have acted as great barriers and provided protection, but the passes provided routes to the migrants and invaders from Central Asia. The sea coast has encouraged contact with people from East and Southeast Asia, Europe and Africa. Navigation technology helped European countries to colonise a number of countries of Asia and Africa, including India as they got accessibility through oceans. **The geographical factors have modified the course of history in different parts of the world.**



Major approaches to study Geography:

- **Systematic Approach:** It was introduced by **Alexander Von Humboldt**, a German geographer (1769-1859). It examines **phenomena globally to identify spatial patterns and typologies**, such as various types of natural vegetation in the world.
- **Regional Approach:** It was developed by another German geographer **Karl Ritter** (1779-1859). In this approach, the **world is divided into regions** at different hierarchical levels and then all the **geographical phenomena in a particular region are studied**. These regions may be natural, political or designated region.

Branches of the Geography:

Physical Geography:

- **Geomorphology** is the study of **landforms**, their evolution, and related processes.
- **Climatology** is the study of structure of **atmosphere** and elements of weather and climates and climatic types and regions.
- **Hydrology** studies the realm of **water over the surface** of the earth including oceans, lakes, rivers and other water bodies and its effect on different life forms including human life and their activities.
- **Soil Geography** aims to study the processes of **soil** formation, soil types, their fertility status, distribution, and use.

Human Geography:

- **Social/Cultural Geography** is the **study of society** and its spatial dynamics as well as the cultural elements contributed by the society.
- **Population and Settlement Geography (Rural and Urban)** studies **population** growth, distribution, density, sex ratio, migration and occupational structure etc. Settlement geography studies the characteristics of rural and urban settlements.
- **Economic Geography** studies **economic activities** of the people including agriculture, industry, tourism, trade, and transport, infrastructure and services, etc.
- **Historical Geography** studies how space is organized over time, considering past experiences and temporal changes in geographical features.
- **Political Geography** examines space through **political events**, focusing on boundaries, space relations between political units, constituency delimitation, election scenarios, and theoretical frameworks for understanding political behavior.

Biogeography: The interface between physical geography and human geography has led to the development of Biogeography which includes:

- **Plant Geography** studies the spatial pattern of natural vegetation in their habitats.
- **Zoo Geography** studies the spatial patterns and geographic characteristics of animals and their habitats.
- **Ecology /Ecosystem** deals with the scientific study of the habitat's characteristic of species.
- **Environmental Geography** concerns world over leading to the realisation of environmental problems such as land gradation, pollution and concerns for conservation has resulted in the introduction of this new branch in geography.

Earth's Shape:

For centuries, the question of Earth's shape puzzled humanity. Ancient civilizations often held the belief that the Earth was flat. Ancient Greek philosophers like **Aristotle** (384-322 BCE) provided arguments for a **spherical Earth** (perfect sphere) based on observations of ships disappearing hull first over the horizon and the Earth casting a round shadow on the moon during a lunar eclipse.

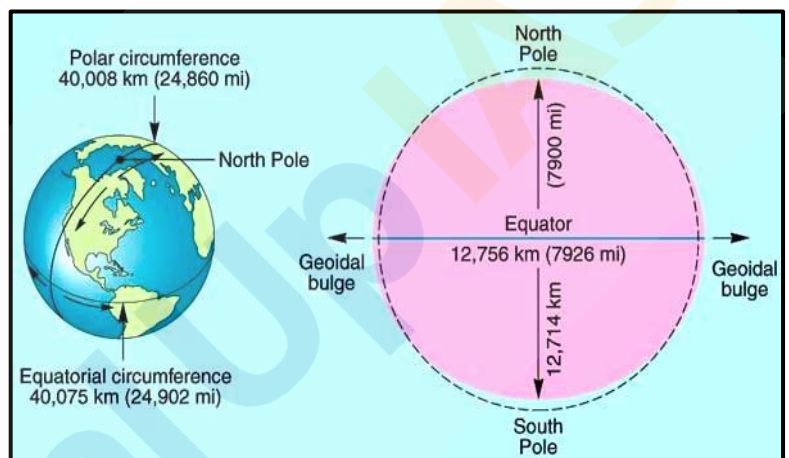
In 1697, **Sir Isaac Newton** challenged the notion of a perfectly round Earth. He proposed that Earth is actually an **oblate spheroid**—a sphere flattened at the poles and bulging at the equator. Newton theorized that **Earth's rotation** causes it to **bulge** due to the **centrifugal force** pushing outward at the **equator**.

Modern '**geodesy**' helps us to understand Earth's shape with even greater precision. Today, the Earth is considered as '**Geoid**'. The geoid means that the shape of the Earth is uniquely '**Earth-shaped**.'

- Slightly flattened Polar diameter = **12714 km**
- Slightly bulging Equatorial diameter = **12756 km**

The **difference** between the **Equatorial Diameter** and the **Polar Diameter** is about **42 km** and in terms of **percentage change**, the flattening or oblateness of the Earth is approximately **0.34%**. This percentage gives an indication of how much the Earth deviates from a perfect sphere.

While the Earth is not a perfect sphere, its overall shape is very close to being spherical. For most practical purposes, such as navigation, surveying, and general geographic calculations, treating the Earth as a sphere is sufficiently accurate. The difference between the Earth's actual shape (an oblate spheroid) and a perfect sphere is very small relative to the Earth's size, making the spherical model quite useful and accurate for many applications.



'Geodesy' is the science of accurately measuring and understanding the Earth's **geometric shape, orientation in space, and gravity field**.

Latitudes and Longitudes:

The fundamental of geographical analysis is the development of a comprehensive and logical framework to **establish the accurate location** of any spot on the Earth's surface.

The **latitudes** and **longitudes** are commonly referred to as geographical coordinates as they provide systematic network of lines upon which the position of various surface features of the earth, can be represented. With the help of these coordinates, location, distance and direction of various points can be easily determined.

Although an infinite number of parallels and meridians may be drawn on a globe, only a selected number of them are usually drawn on a map. Latitudes and longitudes are **measured in degrees (°)** because they represent angular distances. Each degree is further divided into **60 minutes (')** and each minute into **60 seconds (")**.

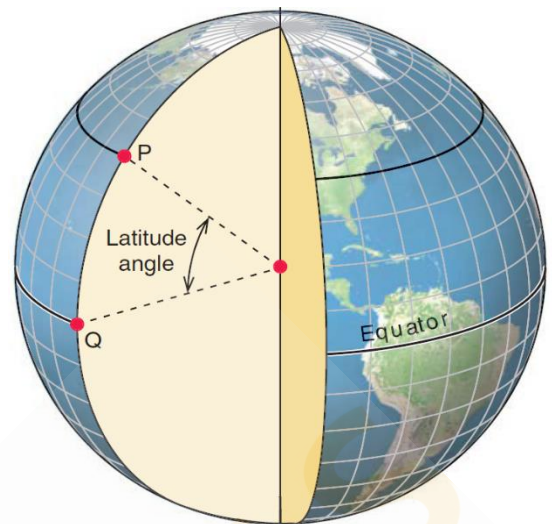


Latitudes:

Latitude is the **angular distance of a point on the Earth's surface** (north or south of the equator), measured in degrees **from the center of the Earth**. Lines joining places with the same latitudes are called '**Parallels.**'

In other words, latitudes are horizontal lines running **parallel to the equator**. The value of **equator is 0°** and the latitude of the **poles are 90°N and 90°S** . The distance between two latitudes is approximately **111 km**.

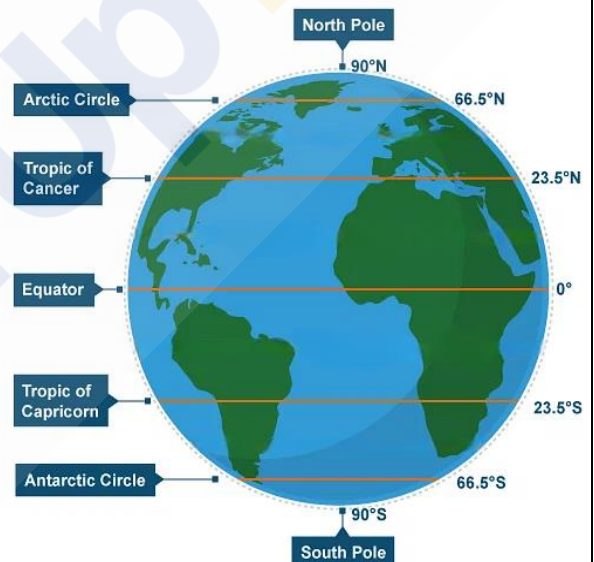
If the latitudes are drawn at an interval of 1 degree, then there will be 89 latitude lines in the northern hemispheres and 89 latitude lines in the southern hemispheres. The total number of parallels thus drawn, including the equator, will be **179**.



Note:- North pole and South Pole are points on the globe and are not drawn as latitudes.

Important Parallels of Latitudes:

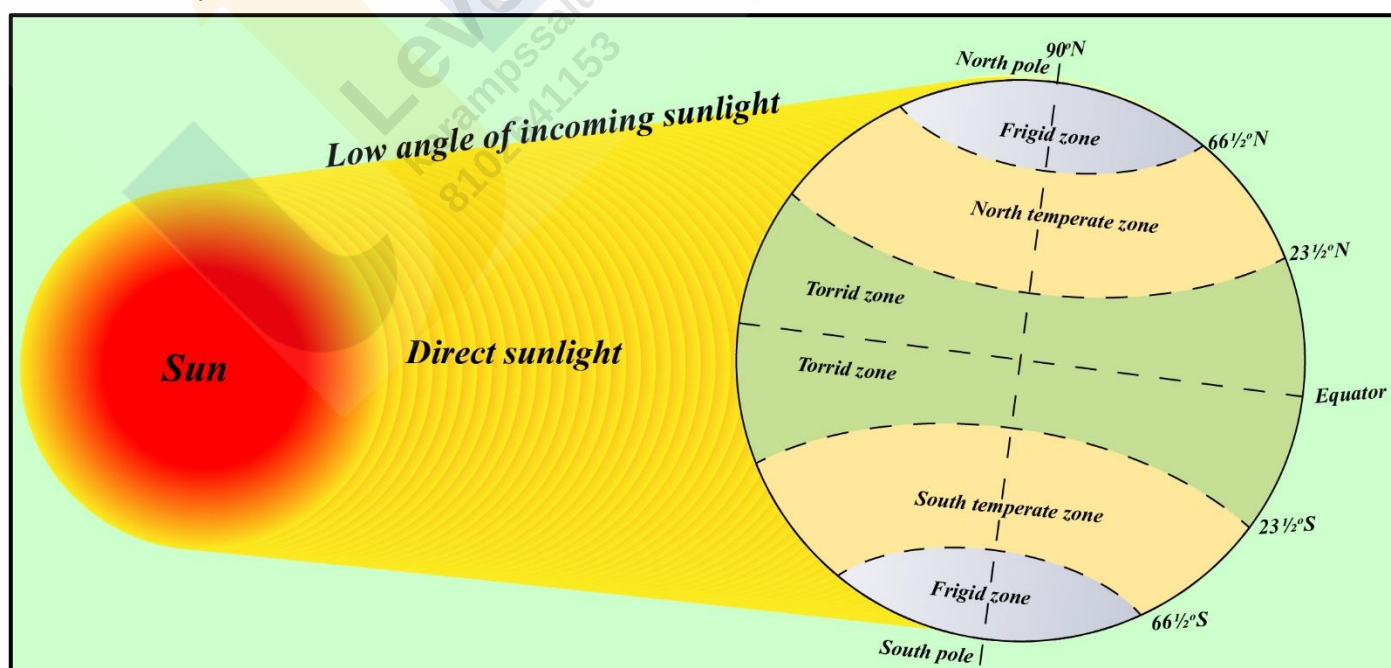
- Equator (0°):** It is an imaginary line that **divides the Earth** into the **Northern and Southern Hemispheres**. It is equidistant from the North and South Poles and represents the point where the **Earth's circumference is the largest (longest line of latitude)**. It covers 40075km out of which 78.8% covers the water area while 21.3% covers the surface area.
 - The Equator passes through 13 countries** - Ecuador, Colombia, Brazil, Sao Tome & Principe, Gabon, Republic of the Congo, Democratic Republic of the Congo, Uganda, Kenya, Somalia, Maldives, Indonesia and Kiribati.
- Tropic of Cancer (23.5°N):** It is the **northernmost latitude** at which the sun can appear **directly overhead at noon** at least once a year (21st June).
 - The Tropic of Cancer passes through 17 countries** - Mexico, Bahamas, Western Sahara, Mauritania, Mali, Algeria, Niger, Libya, Egypt, Saudi Arabia, UAE, Oman, India, Bangladesh, Myanmar, China and Taiwan.
- Tropic of Capricorn (23.5°S):** It is the **southernmost latitude** at which the sun can appear **directly overhead at noon** at least once a year (21st December).
 - The Tropic of Capricorn passes through 10 countries** - Chile, Argentina, Paraguay, Brazil, Namibia, Botswana, South Africa, Mozambique, Madagascar and Australia.
- Arctic Circle (66.5°N):** It marks the southernmost boundary of the Arctic, where the sun can stay **above the horizon** (sun does not set) for 24 continuous hours during the **summer solstice** or be **below the horizon** (sun does not rise) for 24 hours during the **winter solstice**. This phenomenon is known as the Midnight Sun and Polar Night, respectively.
 - The Arctic Circle passes through 8 countries** - Norway, Sweden, Finland, Denmark, Iceland, United Kingdom, Ireland and Canada.



- **Antarctic Circle (66.5° S):** It marks the northernmost boundary of Antarctica and experiences **similar seasonal variations in sunlight as the Arctic Circle**, with extended periods of daylight in summer and darkness in winter.
- **North Pole (90° N):** The North Pole is the **northernmost point** on Earth. It is a point where **all longitudinal lines converge**. The region experiences **six months of continuous daylight** during the **Arctic summer** and **six months of continuous darkness** during the **Arctic winter**.
- **South Pole (90° S):** The South Pole is the **southernmost point** on Earth. Like the North Pole, all longitudinal lines converge here. The South Pole experiences **six months of continuous daylight** during the **Antarctic summer** and **six months of continuous darkness** during the **Antarctic winter**.

Heat Zones of the Earth:

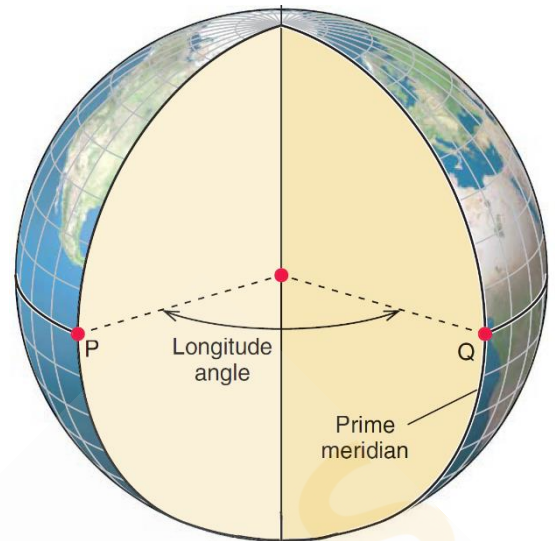
- **Torrid Zone:** This zone, also known as the **tropics**, is the **hottest regions** on Earth. It lies between the **Tropic of Cancer (23.5° N)** and the **Tropic of Capricorn (23.5° S)**. The mid-day sun is **exactly overhead at least once** a year on all latitudes, resulting in consistently high temperatures. This region experiences **minimal variation** in day length and temperature throughout the year.
- **Temperate Zones:** There are two temperate zones - **Northern Temperate Zone** (Between the Tropic of Cancer (23.5° N) and the Arctic Circle (66.5° N)) and **Southern Temperate Zone** (Between the Tropic of Capricorn (23.5° S) and the Antarctic Circle (66.5° S)). Due to the tepid latitude, the midday **sun never shines overhead** on any latitude. The climate in the temperate zones is characterized by **four distinct seasons: spring, summer, autumn, and winter**. Temperatures range from **warm summers and cool winters**, with a moderate level of precipitation spread throughout the year.
- **Frigid Zones:** This zone, also known as the **polar zones**, are the **coldest regions** on Earth. There are two frigid zones - **Arctic Zone** (Extends from the Arctic Circle (66.5° N) to the North Pole (90° N)) and **Antarctic Zone** (Extends from the Antarctic Circle (66.5° S) to the South Pole (90° S)). These zones receive **very little direct sunlight**, and during the winter months, they can experience continuous darkness (polar night), while in the summer months, they can experience continuous daylight (midnight sun). **Winters are long and severe**, while **summers are short** and relatively cool.



Longitudes:

Longitudes, also known as **meridians**, are imaginary lines that run from the North Pole to the South Pole. They measure the **angular distance east or west of the Prime Meridian** (0° longitude). The prime meridian, along with its opposite side at 180 degrees longitude (the anti-meridian), divides the Earth into two hemispheres - **the Eastern Hemisphere** and **the Western Hemisphere**.

Locations east of the Prime Meridian have east longitudes, and locations west have west longitudes. Longitudes are measured in degrees, ranging from **0° to 180° east** and **0° to 180° west**. These lines are used in conjunction with latitudes to determine precise locations on the Earth's surface.



Note:- The word 'meridian' was given by the Romans to refer to the 12-noon local time, which is the moment when the Sun is highest in the sky.

Longitude and Time:

Since the earth makes one complete revolution of **360° in one day** or **24 hours**, it passes through **15° in one hour** or **1° in 4 minutes**.

The Earth is divided into 360 degrees of longitude, **each degree of longitude corresponds to a 4-minute time difference**. This means that places **15 degrees east (or west)** of another location will have a time difference of **1 hour** ($15 \times 4 = 60$ minutes).

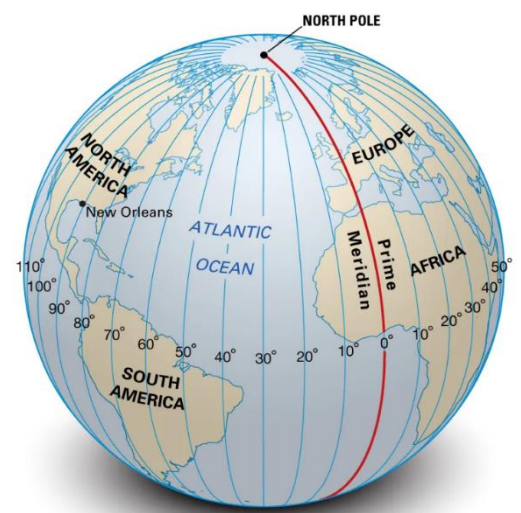
Now, as the **earth rotates from west to east**, so every **15° we go eastwards**, local time is **advanced by 1 hour**. Con-versely, **if we go westwards**, local time is **retarded by 1 hour**.

Standard Time and Time Zones:

Before the establishment of standardized time zones, each location followed its own "**local mean time**" based on the position of the sun in the sky. Noon occurred when the sun reached its highest point. This system worked well locally but created confusion for or global communication, travel, and commerce. Adding to the chaos, different countries used various prime meridians as a reference point for longitude. This made it difficult to accurately determine the difference in local times between geographically distant locations.

By the late 19th century, the need for a standardized system of timekeeping became evident. In **1884**, the United States hosted the '**International Meridian Conference**' in **Washington D.C.** Representatives from 25 nations came together to address this critical issue.

A key outcome of the conference was the adoption of the **Greenwich Mean Time (GMT)** as the global standard for timekeeping. **The Royal Observatory in Greenwich, England**, was designated as the reference point for **zero degrees longitude**, establishing the **Greenwich Meridian** as the **Prime meridian**. This provided a common starting point for calculating longitude and time differences worldwide.



The world was divided into 24 time zones, each differing by one hour from its neighbours. This system ensured that locations within a zone would share the same standard time, simplifying timekeeping for everyday life, travel, and communication.

The Greenwich meridian became the center of a time zone that extends 7.30' of longitude both to the west and east of the Greenwich meridian. Similarly, the meridians that are multiples of 15° (15°, 30°, 45° ...) both east or west of Greenwich meridian were set as **standard meridians** for the 23 other time zones, each of which is 15° of longitude.

12 zones of the east of Greenwich meridian were designated to be ahead of the time at Greenwich by 1 hour per zone. Similarly, the 12 zones to the west of Greenwich meridian are behind.

In International waters, these time zones are shown exactly 7.30' to the east and 7.30' to the west of standard meridian. Over land areas, however, the actual eastern and western boundaries of time zones vary to coincide with appropriate political and economic constraints.

To avoid confusion, a system of standard time is observed by all countries, and such a time is called **Standard Time**. The meridian on which it is based is called '**Standard Meridian**' (Central Meridian).

The Standard meridian is generally chosen by a country with the following norms in mind:

- Central location
- Closeness to some important place.
- It is multiples of 7°30' of longitude so that the standard time differs from Greenwich Mean Time (GMT) by a multiple of 30 minutes (half an hour).

Countries with vast longitudinal extent generally go for **more than one time zone**. Each time zone of approximately 15° of longitudinal extent. **Russia has 11 time zones**. **France**, including its overseas territories, has the **most time zones** with **12** (13 including its claim in Antarctica) in the world.

Note:- The Indian Government has accepted the meridian of 82.5° east for the standard time which is 5 hours 30 mins, ahead of Greenwich Mean Time.

International Date Line (IDL):

The International Date Line (IDL) was **established during the 1884 International Meridian Conference** and is an imaginary line that generally follows the **180th meridian** in the **Pacific Ocean**.

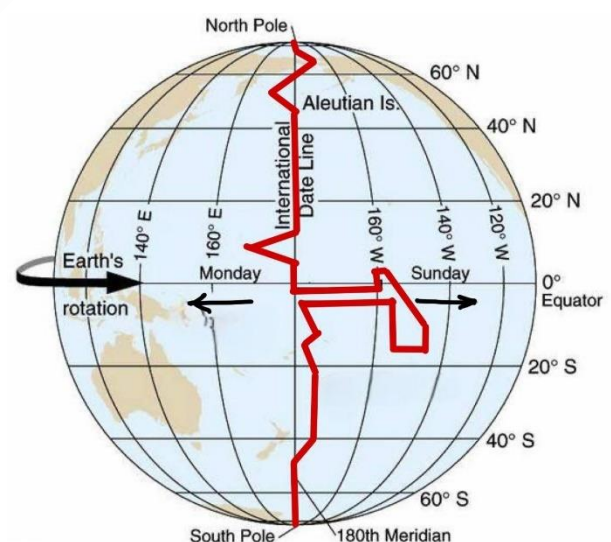
However, this line is **not straight** and **deviate from the 180° meridian at certain points**.

The deviations **enable places of the same economic and political affiliations to share the same date or time**. Otherwise, if the IDL were a straight line, it would divide certain landmasses into two parts with two different dates on the same day.

The IDL serves as the "**line of demarcation**" where the **date changes by one day when crossed**. This adjustment is necessary due to the Earth's rotation and the way time zones are structured.

Crossing the IDL from **west to east** results in "**gaining**" a day and, while crossing the IDL from **east to west** results in "**losing**" a day.

It is interesting to note that **180°E** and **180°W** meridians are on the same line, which is **International Date Line (IDL)**.



Daylight Saving Time (DST):

Daylight Saving Time (DST) is the practice of **setting the clocks forward by one hour during the warmer months** to extend evening daylight. This practice aims to make **better use of daylight** during the longer days of spring, summer, and early autumn.

The concept of DST was first proposed by **Benjamin Franklin** in **1784** in a satirical letter suggesting that people could save candles by getting up earlier to use morning daylight. It was **first implemented during World War I by Germany and its allies** in 1916 to conserve energy.

Mechanism:

- **Spring Forward:** Clocks are set **forward by one hour** in the **spring**, typically around **March or April**, marking the start of DST.
- **Fall Back:** Clocks are set **back by one hour** in the **fall**, usually in **October or November**, ending DST.
- **Example:** In the **United States**, **DST begins** on the **second Sunday in March**, when clocks are set forward from 2:00 AM to 3:00 AM. **DST ends** on the **first Sunday in November**, when clocks are set back from 2:00 AM to 1:00 AM.

Advantages:

- **Energy Conservation** - Reduces artificial lighting and heating.
- **Extended Daylight** - Promotes outdoor activities and boosts retail business.
- **Safety** - Decreases traffic accidents and certain crimes.

Challenges:

- **Health Impacts** - Disrupts sleep patterns and causes fatigue.
- **Limited Energy Savings** - Mixed evidence on actual energy savings.
- **Inconvenience** - Clock adjustments are inconvenient and cause scheduling confusion.

Does India Need Daylight Saving Time (DST)?

Around 30° of longitudinal extent of India (68° 7'E and 97° 25'E) causes significant differences in sunrise and sunset times, notably affecting the **Northeast** where the **sun rises around 4 a.m. in summer and sets by 4 p.m. in winter**. This leads to calls for two time zones or the adoption of Daylight-Saving Time (DST) to better align with natural daylight hours. **India currently does not observe Daylight Saving Time (DST).**

Arguments against DST in India:

- **Limited Impact:** India is a vast country located close to the equator. Daylight hours don't vary significantly throughout the year, minimizing the potential benefit of "borrowing" an hour of sunlight in the evenings.
- **Disruption of Routines:** The twice-yearly clock adjustments could disrupt sleep patterns and daily routines for a large population, potentially impacting productivity and well-being.
- **Energy Savings Debate:** Studies on the energy-saving benefits of DST in countries with larger seasonal variations in daylight hours are inconclusive. The impact on energy consumption in India might be negligible.
- **Logistical Challenges:** Implementing DST nationwide would require significant logistical coordination across various sectors, including transportation, communication, and IT systems.

Arguments against DST in India:

- **Potential Benefits in Specific Regions:** A separate time zone for the Northeast could spur regional development by facilitating more efficient use of daylight hours for economic activities. This could benefit sectors like agriculture, tourism, and local industries, which rely heavily on daylight for operational efficiency.
- **Energy Savings:** Establishing two time zones could lead to significant energy savings, estimated at about 20 million kWh annually. By better matching daylight hours with operational hours, especially in the Northeast, there would be reduced electricity consumption for lighting and heating during early morning and late afternoon hours.
- **Improved Safety and Health:** Aligning working hours with daylight can reduce risks associated with early morning commutes and late evening activities. This includes lower risks of road accidents due to better visibility during commuting hours and potentially reducing certain crimes associated with darker evenings.

Alternative Proposal:

- Researchers from the **National Institute of Advanced Studies** in Bangalore and scientists at **CSIR-NPL** propose **advancing Indian Standard Time (IST) by half an hour to make it six hours ahead of Universal Coordinated Time (UTC)**. This adjustment would shift the reference longitude from 82.5 degrees East to **90 degrees East**, located along the West Bengal-Assam border.

The Indian government has not shown any recent interest in implementing DST. The potential drawbacks seem to outweigh the limited benefits in the context of India's geographical location and daylight patterns. The country's focus on energy conservation is likely to target other areas with potentially greater impact.

Chaibagaan Time

150 years ago, British colonialists introduced “**chaibagaan time**” or “**bagaan time**”, a time schedule observed by tea planters, which was **one hour ahead of IST**. This was done **to improve productivity by optimizing the usage of daytime**. After Independence, Assam, along with the rest of India, has been following IST for the past 66 years. The administration of the Indian state of Assam now wants to change its time zone back to Chaibagaan time to conserve energy and improve productivity. Indian government didn't accept to such a proposal.

Some important concepts:

Circadian Rhythms:

Circadian rhythms refer to the **natural, internal process that regulates the sleep-wake cycle** and repeats roughly every **24 hours**. These rhythms are influenced primarily by **light and darkness**, which help synchronize our **biological clock** with the day-night cycle.

Jet Lag:

Jet lag occurs when **rapid travel across multiple time zones** disrupts our circadian rhythms. This disruption can lead to symptoms like **fatigue, difficulty concentrating, and disturbed sleep patterns**. The severity of jet lag symptoms depends on factors such as the number of time zones crossed and the direction of travel.

Eastward travel typically **causes more significant jet lag** because it requires adjusting to shorter days, whereas westward travel allows for longer days, easing the adjustment.

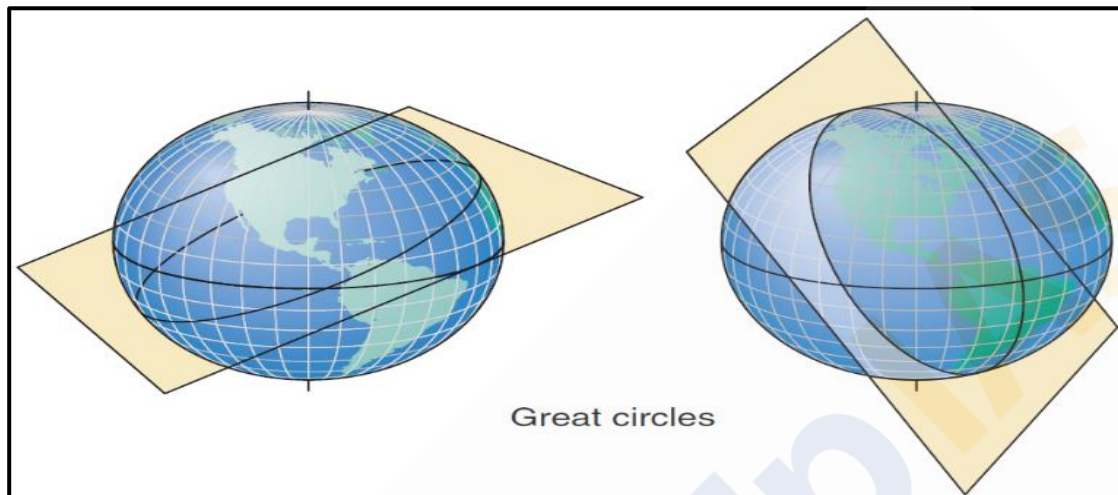
Great Circle:

Any plane that is passed through the center of a sphere bisects that sphere and creates a great circle where it intersects the surface of the sphere.

A great circle is the largest possible circle that can be drawn around a sphere.

Of all the parallels (latitudes), equator alone makes a great circle. All other parallels make small circles.

Each meridian is a 'semi-great circle'. Taking the opposite side of the other part of meridian makes it a 'great circle'.



Circle of Illumination:

The sun illuminates one half of the earth at any given moment. The "Circle of Illumination" refers to the **dynamic line that divides the Earth into two hemispheres - the hemisphere experiencing daylight (day) and the hemisphere experiencing darkness (night) at any given moment.** It is a great circle that divides the globe at all times into a **sunlit hemisphere** and a **shadowed hemisphere**.

This **line shifts continuously due to the Earth's rotation on its axis**, completing one full rotation approximately every 24 hours. As the Earth rotates eastward, areas to the west of the line of illumination move into darkness as they face away from the Sun, while areas to the east move into daylight as they face towards it.

This phenomenon is crucial for understanding how different parts of the world experience varying lengths of daylight and darkness throughout the year, influencing seasonal patterns and daily activities across the globe.

